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# Running off-ball metrics

Definitions of the off-ball running metrics

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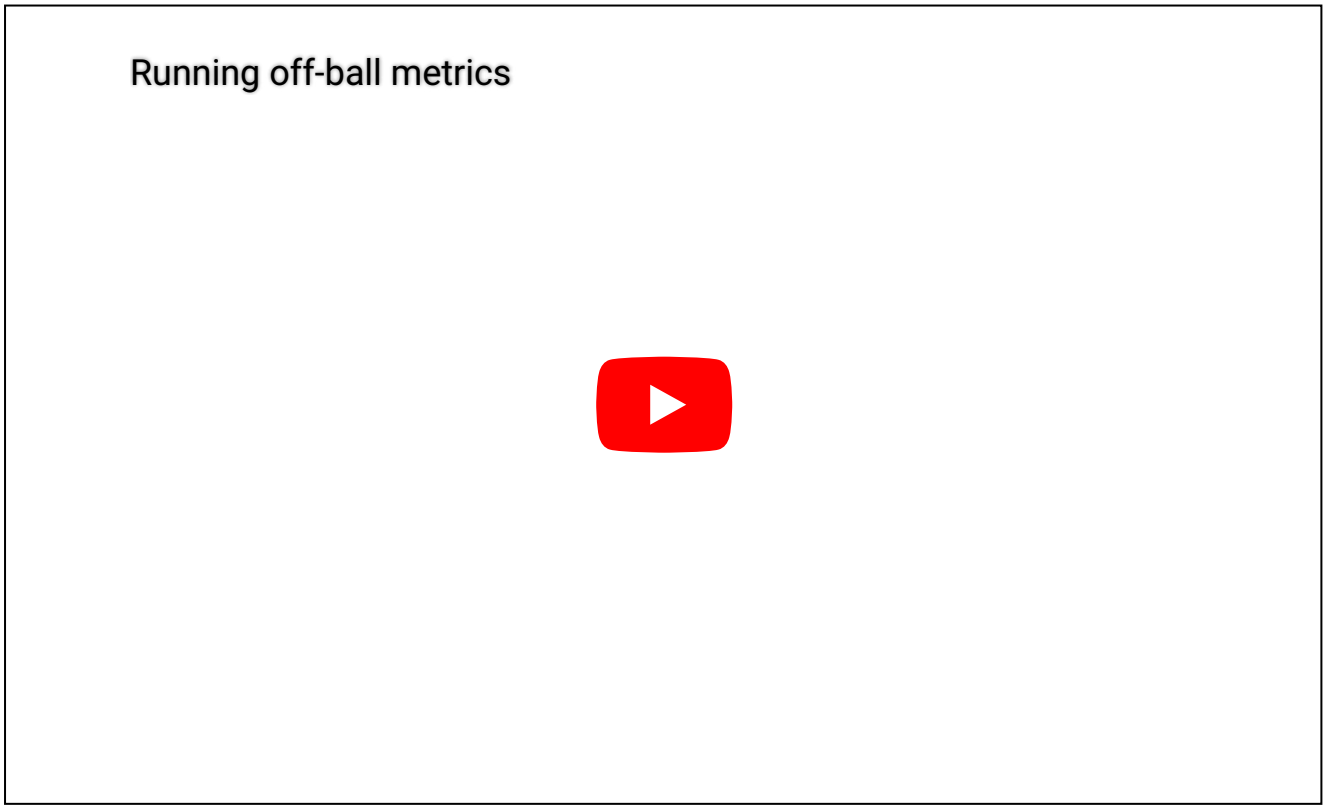
The off-ball running metrics give insights related to the volume, the type and the quality of the off-ball runs players are performing when their team is in possession.

We start with a simple set of metrics to address questions like “how often is this player performing runs in behind the defense” or “how often is he served while doing so”.

We then add an extra layer of context with metrics related to the dangerousness of the runs performed by the players in order to answer questions like “how many chances is this player creating with his runs?”.



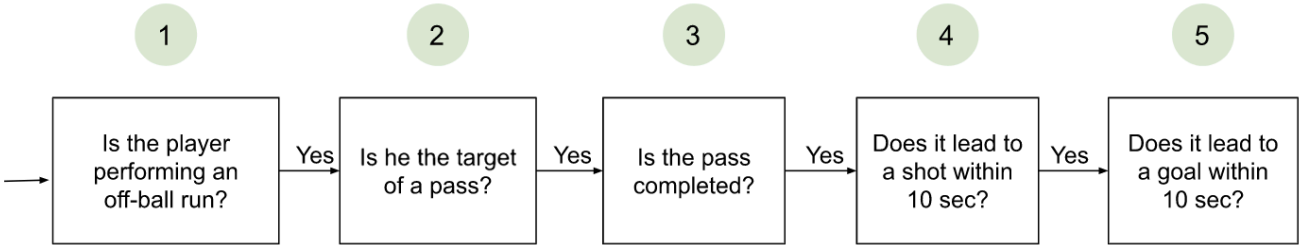
For further details on the metrics, read the next sections.

# A recurring situation

All the metrics relate to a simple situation that occurs repeatedly during every match.

Considering a player from the team in possession, we look at the following elements:

1. Is he performing an off-ball run?
2. If yes, is he the target of a pass from the player who has the ball?
3. If yes, does he receive the ball?
4. If yes, does the completed pass lead to a shot within 10 seconds?
5. If yes, does the completed pass  lead to a goal within 10 seconds?




Off-ball run situation

# Describing what we see

To capture a player's behaviour in such situations, we count how many times he reaches each step of the sequence over multiple situations.

We use the definition of “Run” from this article.

All the metrics in this article can be broken down per run type, channel and third of the pitch. They can also be averaged/normalised per match, per 100 runs or per 30 min tip.

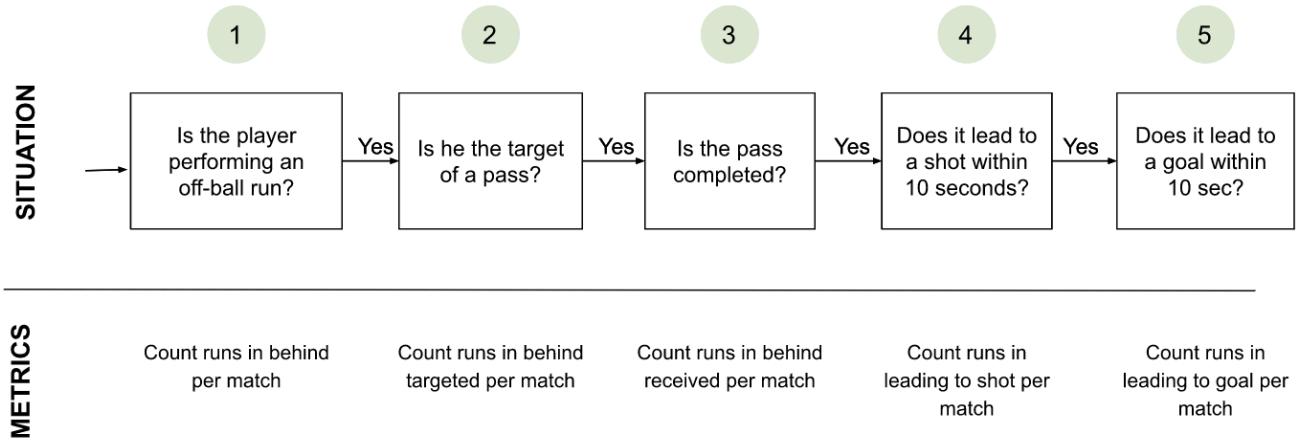
For further details

- on the run types, read this article.
- on the pitch area, read this article.
- on how averages are calculated, read this article.

> For illustrative purposes, in this article, we consider runs in behind on the whole pitch, averaged per match.

The first set of metrics is composed of the following 5 metrics:

1. **Count runs in behind per match**: Number of runs in behind performed by the player.
2. **Count runs in behind targeted per match**: Number of runs in behind performed by the player where he was the target of a pass from the player in possession of the ball.
3. **Count runs in behind received per match**: Number of runs in behind performed by the player where he received the ball.
4. **Count runs in behind leading to shot per match**: Number of runs in behind performed by the player where he received the ball and it led to a shot in the 10 seconds following the pass from the passer.
5. **Count runs in behind leading to goal per match**: Number of runs in behind performed by the player where he received the ball and it led to a goal in the 10 seconds following the pass from the passer.




5 metrics to describe what happens when a player perform an off-ball run

We provide one additional metric to help assess the level of confidence one can have in the metrics values.

**Count runs in behind in sample**: Number of runs in behind taken into account to compute the metrics (it is particularly important when calculating average as the bigger the sample size the more trustworthy the value is).

> For instance, if one considers a player who played 3 matches since the start of the season where he performed 5, 10 and 6 runs in behind, its stats since the start of the season will be:
>
> - Count runs in behind per match equals 7 = (10 + 5 + 6) : 3
> - Count runs in behind in sample equals 21

# Focus on dangerousness

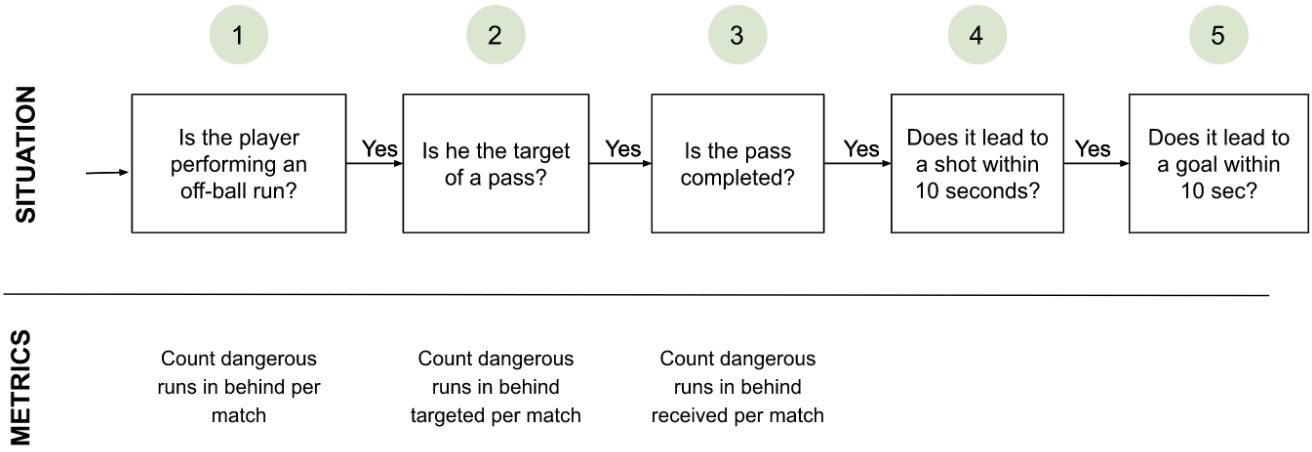
One may want to go further and start evaluating the potential impact of the runs a player is performing.

We propose to address this question by focusing on how dangerous these runs are.

> **Reminder**
> A dangerous run is a run with at least 2% chance to lead to a goal within 10 seconds if the player performing the run was to be served.
> The evaluation of the chance of a run leading to a goal is done using SkillCorner Threat Model.
>
> For further details read this article.

We compute the same metrics as those presented above except we filter them on dangerous runs:

- **Count dangerous runs in behind per match:** Number of times the player performed a dangerous run in behind, meaning the run threat was superior to 2% (which is equivalent to saying that there was at least 2% chance that a goal would have been scored within 10 seconds if the runner received the ball).
- **Count dangerous runs in behind targeted per match:** Number of times the player performed a dangerous run in behind and he was the target of a pass from the player in possession of the ball.
- **Count dangerous runs in behind completed per match:** Number of times the player performed a dangerous run in behind and he received the ball.




Dangerous runs focus

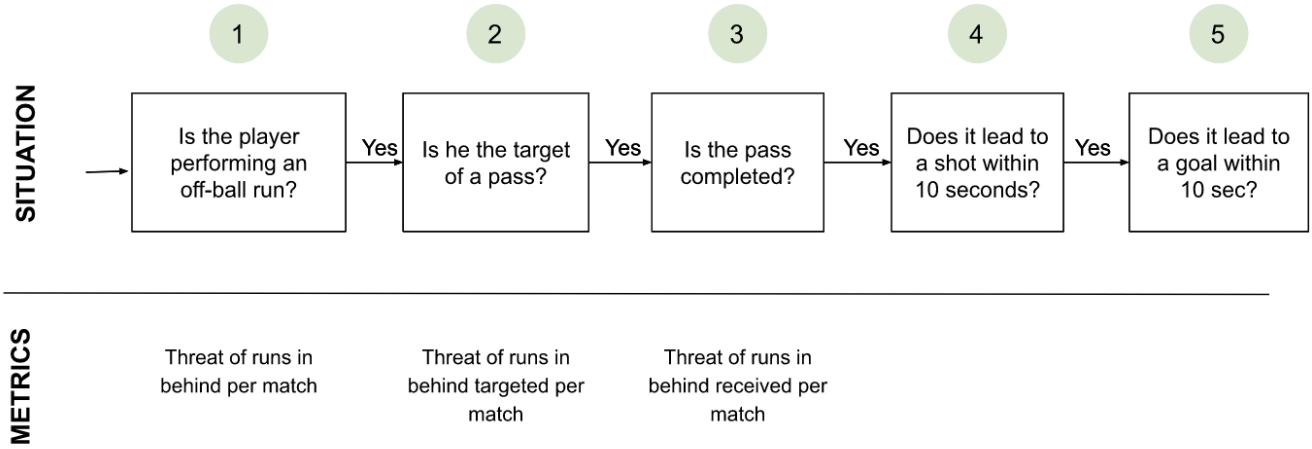
# Chances of scoring created

Using the SkillCorner Threat Model, we evaluate the threat associated with the off-ball runs performed by the player.

The Threat associated with a given off-ball run corresponds to the chance of a goal being scored by this player's team within 10 seconds if he had received the ball.

The model simulates what would happen after the player received the ball using average players.

1. **Threat of runs in behind per match**: Sum of the run threats associated to the runs in behind performed by the player. It corresponds to the number of goals that would have been scored by an average player who would have received the ball while performing the same runs in behind.
2. **Threat of runs in behind targeted per match**: Sum of the run threats associated to the runs in behind performed by the player where he was the target of a pass from the player in possession of the ball.
3. **Threat of runs in behind receveid per match**: Sum of the run threats associated to the runs in behind performed by the player where he received the ball.

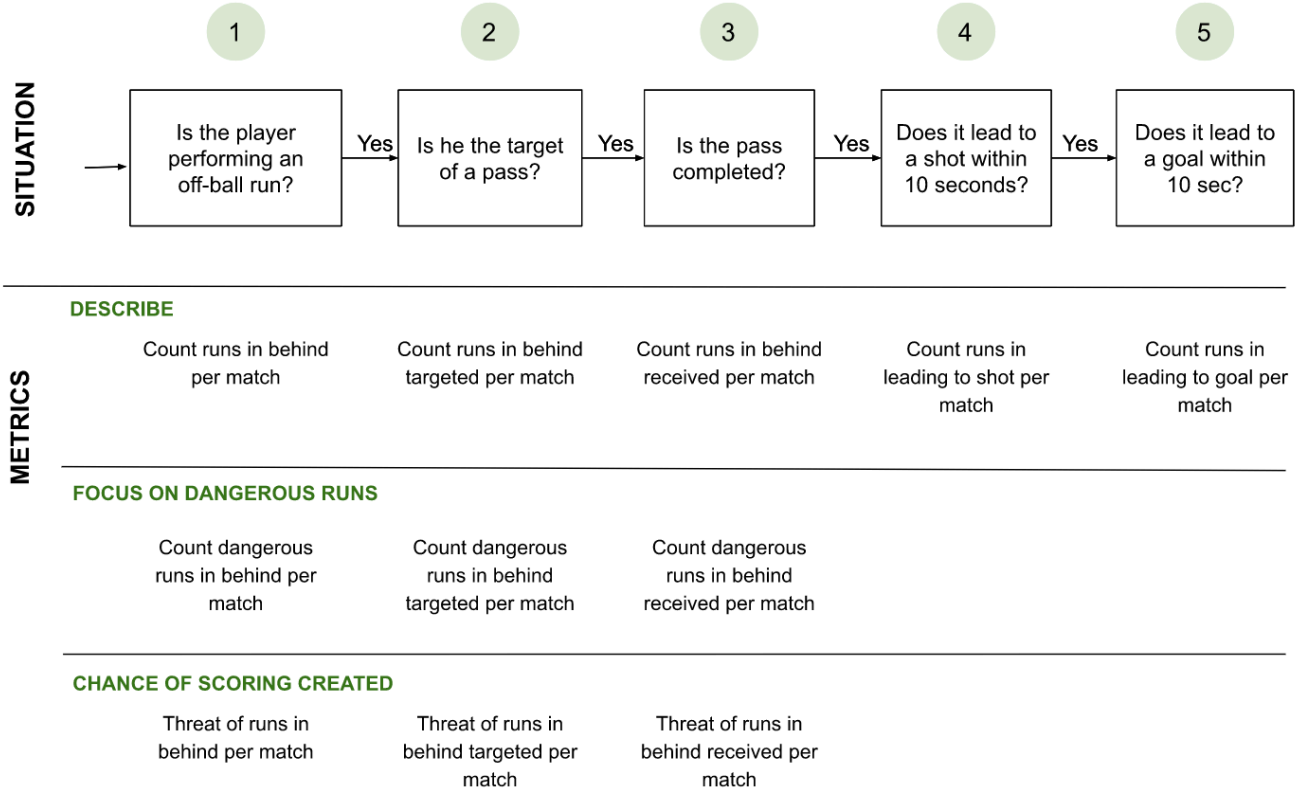



Assessing the chances of a goal being scored

# The off-ball runs matrix

The main metrics can be grouped under one matrix, the off-ball runs matrix.

It synthesises the relation between the situation we consider and the metrics.




The off-ball running matrix

In order to analyse player performance, one can look at the what happen when going through the matrix horizontally (e.g. by comparing how many runs a player did Vs how many time he was served) or looking at it vertically e.g. comparing how many runs a player did compared to the number of dangerous runs he did).

# FAQ

**How do you deal with consecutive runs?**

Scenario 1: The player A is performing a run while the player in possession B passes the ball to another player C.

In this case, only one run will be credited to the runner, and this run will be associated with only one possession (either player B possession or player C's).

Scenario 2: The player is performing a run, he stops and starts again.

In this case we would credit two runs to the player.

**How do you deal with a run that goes through different run types?**

If a player goes through different run types while meeting the requirements from the run definition (see this article) then we account credit the player with only one run of a given type.

The run type is determined based on the last 4 seconds of the run.

**At which moment of the runs are the threats collected?**

The threat of an off-ball run corresponds to the average of the threat over the last second of the run.

**How do you split the data by third and channel?**

The third and the channel of a run correspond to the location where an off-ball run **ends**.

DID THIS ANSWER YOUR QUESTION?

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